

The exchange move is a re-timing, not a hop: a TASEP reading of Theorem 4.1, and the death of a conjecture

Clio

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Author: Clio

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Recipient: Robin Langer, cc Rick (grandpa-rick) **Session:** PROVE c2, Q227 / Q221

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Abstract

DREAM c1 of 2026-09-22 conjectured (conjecture (C)) that the exchange-move set of [ClioExch, Thm. 4.1] is “the family of local moves that a TASEP exclusion rule allows”, and offered a cheap falsifiable test: that the number of exchange moves at a length-additive pair (S, T) should equal the number of available hops of the corresponding TASEP state. **The test fails, and (C) is false as stated.** The count of available hops is not even an invariant of $v = u_S u_T$, whereas the count of exchange moves is ([ClioExch, Thm. 3.1], localisation); so no such identity can hold. Numerically the two agree on 2111 of 9168 realisable configurations for $N \leq 7$.

What replaces it is sharper than what was conjectured, and is proved here. Under Lam’s nilCoxeter action [Lam, 1.1700–1721] a length-additive pair (S, T) acting on a cylindric shape is exactly *two time-steps of a TASEP on a ring of N sites*, and S and T are the sets of *arcs* crossed at step 2 and step 1 respectively. On that locus we prove (Theorem 3.1) that Theorem 4.1’s letter e is *always* the bottom m of its run — never an interior letter — so that every exchange move is

$$S \mapsto S \setminus \{m\}, \quad T \mapsto T \cup \{m\},$$

and (Theorem 4.1) that this advances the intermediate configuration by exactly one TASEP hop: the particle crosses its first arc one time-step earlier. The endpoints of the trajectory, and the multiset of arcs crossed, are untouched. **The exchange move is a re-timing of a fixed trajectory, not a hop of a state**, and that is precisely why no single-state hop count can measure it.

The locus is exactly the 321-avoiding one, and the dichotomy is sharp: off it, $e > m$ occurs in 2460 of 6270 pairs and the move has no trajectory reading at all. We also report a negative that bears on the same dream: the Morse–Schilling excess of [ClioCrystal] is *not* confined off the cylindric locus (574 of the 1001 excess instances for $N \leq 7$ are 321-avoiding), so the strictness proved on 09-22 c1 is not explained by the cylindric/non-cylindric split.

Contents

1	What was asked	1
2	The dictionary: beads, arcs, and two time-steps	2
2.1	Lam’s action is an exclusion rule	2
2.2	What a cyclically decreasing element does to the beads	3
2.3	The locus on which the dictionary exists	4

3	The main theorem: the letter is always the run bottom	4
4	The move is a re-timing	5
5	The death of (C)	6
5.1	What survives, and what the right statement is	6
5.2	A second negative: the Morse–Schilling excess is not the locus	7
6	Computational verification	7
7	Gaps	8

1 What was asked

Let $N \geq 3$ and let $\tilde{S}_N$ be the affine symmetric group, with $S \subsetneq \mathbb{Z}/N$ identified with its N -periodic lift and u_S the unique cyclically decreasing element of support S , $\ell(u_S) = |S|$ [ClioExch, Lem. 2.1]. Call (S, T) *length-additive* if $\ell(u_S u_T) = |S| + |T|$, and write $v = u_S u_T$.

Theorem 1.1 ([ClioExch, Thm. 4.1]). *Let (S, T) be length-additive with $|S| > |T|$. Then there is a run $[m, M]$ of S with $m \notin T$ and an $e \in [m, M]$ with $e + 1 \notin T$; for the first such e in the order $m, m + 1, \dots, M$, putting $S' = S \setminus \{e\}$, $T' = T \cup \{m\}$, one has $u_{S'} u_{T'} = v$ and $\ell(v) = |S'| + |T'|$.*

Call a run $[m, M]$ of S *usable* if $m \notin T$ and $j + 1 \notin T$ for some $j \in [m, M]$. Theorem 1.1 produces one move per usable run; write $\mathcal{E}(S, T)$ for the set of letters e so produced, so $|\mathcal{E}(S, T)| = \#\{\text{usable runs of } S\}$.

The conjecture under test, from `memory/connections/2026-09-22-a-crystal-operator-is-a-canonical-p`

(C) *The Theorem 4.1 exchange move set is the family of local moves that a TASEP exclusion rule allows; Morse–Schilling’s x -pairing is the tie-break that selects one of them. Testable prediction: $|\mathcal{E}(S, T)|$ counts the available hops of the corresponding TASEP state.*

The brief for this session required a dictionary before any count (`dictionary-before-identification`), and required that a mismatch be reported rather than patched. Both requirements are met below: §2 builds the dictionary, §3 proves what is true on it, and §5 reports the failure.

Remark 1.2 (One half of the test was dropped before it was run). As recorded in the WAKE addendum to `questions/2026-09-22-dream-Q221-Q227-ledger.md`, the second half of Q227 — “is $|\bigcup_x \tilde{e}_1^{(x)}(uv)|$ at most $|X(uv)|$?” — follows from the definition alone, since $\tilde{e}_1^{(x)}$ is a function of x and the union is a union of at most $|X(uv)|$ singletons. It is not instrumented here.

2 The dictionary: beads, arcs, and two time-steps

2.1 Lam’s action is an exclusion rule

Definition 2.1 (bead configuration). A *bead configuration* on the ring $\mathbb{Z}/N$ is a set $A \subseteq \mathbb{Z}$ with $A + N = A$ and $\emptyset \neq A/N\mathbb{Z} \subsetneq \mathbb{Z}/N$. We read a bead as a particle and a non-bead as an empty site, and we freely identify A with $A/N\mathbb{Z}$.

A cylindric shape of type (N, m) in the bead model of [ClioCyl, §2] is a strictly increasing $x : \mathbb{Z} \rightarrow \mathbb{Z}$ with $x_{i+m} = x_i + N$; its bead set is $A = \{x_i\}$, which is a bead configuration in the above sense. Adding one box per period to row j is $x_j \mapsto x_j + 1$, legal exactly when $x_j + 1 \notin A$. Lam’s nilCoxeter action [Lam, 1.1700–1721] (“ $u_i \cdot \lambda = s_i \cdot \lambda$ if that adds boxes, else 0”), transported to bead coordinates by the derivation recorded in [Clio321, §6.1] — which fixes the convention without any freedom except the origin of the labelling — reads:

$$u_i \cdot A = \begin{cases} (A \setminus \{i\}) \cup \{i+1\} & \text{if } i \in A, i+1 \notin A, \\ 0 & \text{otherwise.} \end{cases} \quad (1)$$

Observation 2.2. Equation (1) is verbatim the totally asymmetric exclusion rule: *the particle at site i hops one step clockwise if and only if site $i+1$ is empty.* This is not an analogy imported from physics; it is Lam’s box-adding rule written in the coordinates in which a cylindric shape is a particle configuration on a ring. That standard cylindric tableaux are in bijection with sequences of TASEP states is the main theorem of [Eliz-B], whose abstract reads, verbatim, “*We establish bijections between three classes of combinatorial objects . . . lattice walks in simplicial regions . . . , standard cylindric tableaux . . . , and sequences of states in the totally asymmetric simple exclusion process.*”

Remark 2.3 (Which paper supplies what). [Eliz-B] (and its FPSAC 2026 extended abstract [Eliz-A]) supplies the *name* and the wider context: its $\Delta_{d,\ell}$ picture, with $x_i = \alpha_{i-1} - \alpha_i$ [Eliz-A, §4.2], is the gap-vector form of the same ring, and its property (D3) — *every cylindric shape covers and is covered by $c(\rho) = \#\{i : \rho_{i-1} > \rho_i\}$ elements* [Eliz-A, p. 5] — is the statement that the numbers of available forward and backward hops agree. Nothing in §§2–4 *consumes* [Eliz-B]: the proofs run on (1) alone. The bridge from Morse–Schilling’s affine-Stanley setting to Elizalde’s cylindric-tableau setting, which the DREAM note flagged as an open gap, is Lam’s **thm:321** together with its converse, quoted in §2.3; it is closed here for the exchange move, and only for the exchange move.

2.2 What a cyclically decreasing element does to the beads

Write $\text{runs}(S)$ for the set of maximal cyclic runs of $S \subsetneq \mathbb{Z}/N$, each lifted to an interval $[m, M] \subseteq \mathbb{Z}$ with $M - m + 1 \leq |S| \leq N - 1$. The cyclically decreasing reduced word for u_S reads each run downwards, $s_M s_{M-1} \cdots s_m$, runs concatenated in any order; in the product $u_S u_T$ the *rightmost* letter acts first ([Lam, Alternative Definition 2, 1.853–861]: “*the a_1 factor acts first, i.e. is the rightmost*”). Hence in $v = u_S u_T$ the factor u_T acts before u_S .

Lemma 2.4 (block travel). *Let A be a bead configuration and $S \subsetneq \mathbb{Z}/N$. Then $u_S \cdot A \neq 0$ if and only if for every run $[m, M]$ of S*

$$m \in A \quad \text{and} \quad [m+1, M+1] \cap A = \emptyset, \quad (2)$$

and in that case

$$u_S \cdot A = \left(A \setminus \{m : [m, M] \in \text{runs}(S)\} \right) \cup \{M+1 : [m, M] \in \text{runs}(S)\}.$$

That is: for each run, the single bead at its bottom m travels to $M+1$, crossing the arcs $m, m+1, \dots, M$, and the whole interval it crosses must be empty.

Proof. Distinct runs $[m, M]$, $[m', M']$ of S have disjoint blocks $[m, M + 1]$, $[m', M' + 1]$ modulo N : if $M + 1 = m'$ then $m' \notin S$ and $m' \in S$, absurd; and the blocks cannot overlap otherwise because the runs are separated by gaps. The operators for distinct runs therefore act on disjoint sets of sites and commute, so it suffices to treat one run.

For one run the word is $s_M s_{M-1} \cdots s_m$, applied rightmost-first, i.e. we apply $u_m, u_{m+1}, \dots, u_M$ in that order. We claim by induction on $j = m - 1, m, \dots, M$ that after applying $u_m, \dots, u_j$ the configuration is A with the bead at m replaced by a bead at $j + 1$, and that this is nonzero exactly when $m \in A$ and $[m + 1, j + 1] \cap A = \emptyset$. The base $j = m - 1$ is the empty statement. For the step, u_{j+1} requires a bead at $j + 1$ — present, it is the travelling bead — and no bead at $j + 2$ in the *current* configuration. The current configuration differs from A only in that m was removed and $j + 1$ added, and $j + 2 \notin \{m, j + 1\}$: indeed $j + 2$ ranges over $[m + 2, M + 1]$ and $M + 1 - m \leq |S| \leq N - 1$, so $j + 2 \neq m$. Hence the condition is $j + 2 \notin A$. Composing the conditions over $j = m - 1, \dots, M - 1$ gives exactly (2). $\square$

Definition 2.5 (arcs). Label the arc from site i to site $i + 1$ by i . By Lemma 2.4, if $u_S \cdot A \neq 0$ then S is precisely the set of arcs crossed by the beads during the step $A \mapsto u_S \cdot A$, each arc crossed exactly once, each particle crossing a contiguous block of arcs.

So $\ell(u_S) = |S|$ is the total displacement of the step — equivalently, by [ClioCyl, §2], the number of cells of the horizontal strip added — and the *trajectory* associated with a length-additive pair is:

$$A \xrightarrow{\text{step 1, arcs } T} \nu := u_T \cdot A \xrightarrow{\text{step 2, arcs } S} \lambda := u_S \cdot \nu. \quad (3)$$

2.3 The locus on which the dictionary exists

Definition 2.6. A length-additive pair (S, T) is *realisable* if there is a bead configuration A with $u_S \cdot (u_T \cdot A) \neq 0$, i.e. if the trajectory (3) exists.

Proposition 2.7 (the locus is the 321-avoiding one). (S, T) is *realisable* if and only if $v = u_S u_T$ is 321-avoiding.

Proof. $(\Leftarrow)$ is [Lam, thm:321, 1.1844–47] in the form recorded in [Clio321, Thm. 5.2]. $(\Rightarrow)$ is Lam’s converse, [Lam, 1.1871–73], verbatim: “If λ and μ are cylindric shapes satisfying $u_w \cdot \mu = \lambda$ then w is necessarily 321-avoiding.” Independently verified here on all 10729 length-additive pairs for $3 \leq N \leq 7$: agreement 10729/10729 (§6, **V2**). $\square$

3 The main theorem: the letter is always the run bottom

Theorem 3.1. Let A be a bead configuration, let (S, T) be length-additive, and suppose $\nu = u_T \cdot A \neq 0$ and $\lambda = u_S \cdot \nu \neq 0$. Let $[m, M]$ be any run of S . Then

- (a) $m \notin T$;
- (b) either $m + 1 \notin T$, or $[m + 1, M + 1] \subseteq T$;
- (c) the run is usable if and only if $m + 1 \notin T$, and in that case Theorem 1.1’s letter is $e = m$.

Consequently every exchange move of Theorem 1.1 at such an (S, T) is

$$S \mapsto S \setminus \{m\}, \quad T \mapsto T \cup \{m\},$$

for a run bottom m of S with $m + 1 \notin T$, and

$$|\mathcal{E}(S, T)| = \#\{[m, M] \in \text{runs}(S) : m + 1 \notin T\}. \quad (4)$$

Proof. Throughout, Lemma 2.4 is applied twice: to T acting on A , and to S acting on ν . From the second,

$$m \in \nu \quad \text{and} \quad [m + 1, M + 1] \cap \nu = \emptyset. \quad (*)$$

From the first, $\nu = (A \setminus \{\text{run bottoms of } T\}) \cup \{b + 1 : [a, b] \in \text{runs}(T)\}$, and each run $[a, b]$ of T satisfies $a \in A$, $[a + 1, b + 1] \cap A = \emptyset$.

(a). Suppose $m \in T$ and let $[a, b]$ be the run of T containing m . If $m = a$, then the bead at a left, and a receives a bead in ν only if $a = b' + 1$ for some run $[a', b']$ of T ; but $b' + 1 \notin T$ while $a \in T$. So $m = a \notin \nu$, contradicting (*). If $a < m \leq b$, then $m \in [a + 1, b + 1]$ so $m \notin A$, and again $m \in \nu$ would force $m = b' + 1 \notin T$, contradicting $m \in T$. So $m \notin \nu$, again contradicting (*). Hence $m \notin T$.

(b). Suppose $m + 1 \in T$. By (a), $m \notin T$, so the run of T containing $m + 1$ begins at $m + 1$: call it $[m + 1, b]$, with $b \leq m + N - 1$. By Lemma 2.4 the bead travelling along it lands at $b + 1$, so $b + 1 \in \nu$. By (*), $b + 1 \notin [m + 1, M + 1]$; since $b + 1 \geq m + 2$, this forces $b + 1 \geq M + 2$, i.e. $b \geq M + 1$. Hence $[m + 1, M + 1] \subseteq [m + 1, b] \subseteq T$.

(c). By (a) the run passes the first test of usability. If $m + 1 \notin T$, then $j = m$ satisfies $j + 1 \notin T$, so the run is usable and the *first* such j is $e = m$. If $m + 1 \in T$, then by (b) $[m + 1, M + 1] \subseteq T$, so $j + 1 \in T$ for every $j \in [m, M]$ and the run is not usable. Equation (4) follows, since Theorem 1.1 contributes exactly one move per usable run. $\square$

Remark 3.2 (why an interior letter is impossible, in one sentence). In an exclusion process only the *rear* particle of a moving block can be held back: if an interior arc of a particle's journey were re-timed, the particle behind it would have to pass through it. Theorem 3.1(c) is that sentence, proved.

Remark 3.3 (the hypothesis $|S| > |T|$ is not used). Theorem 3.1 places no hypothesis relating $|S|$ and $|T|$. In [ClioExch, Thm. 4.1] the hypothesis $|S| > |T|$ is consumed *only* by the applicability lemma, which guarantees that a usable run *exists*; the move itself needs only a usable run. Comparing against what the theorem guarantees rather than what it constructs is the error recorded in `a-hypothesis-of-my-own-theorem-is-an-unexamined-assumption`, and it is avoided here.

4 The move is a re-timing

Theorem 4.1. *In the situation of Theorem 3.1, let $[m, M]$ be a run of S with $m + 1 \notin T$, and put $T' = T \cup \{m\}$, $S' = S \setminus \{m\}$. Then*

$$\nu' := u_{T'} \cdot A = (\nu \setminus \{m\}) \cup \{m + 1\} \neq 0, \quad u_{S'} \cdot \nu' = \lambda.$$

That is: the exchange move advances the intermediate configuration by exactly one TASEP hop — of the very particle that is about to make the step-2 journey $m \rightarrow M + 1$ — and leaves the endpoints A and λ of the trajectory unchanged. Equivalently, in the language of Definition 2.5, it moves the crossing of the single arc m from step 2 to step 1.

Proof. By Theorem 3.1(a),(c) we have $m \notin T$ and $m + 1 \notin T$. Two facts first.

$m \in A$ or $m = b + 1$ for the run of T ending at $m - 1$. Since $m \in \nu$ by $(*)$, either $m \in A$ and m is not a run bottom of T (which holds, as $m \notin T$), or $m = b + 1$ for some run $[a, b]$ of T , i.e. $b = m - 1 \in T$.

$m + 1 \notin A$. By $(*)$, $m + 1 \notin \nu$. If $m + 1 \in A$ then, ν being obtained from A by deleting run bottoms of T and adding tops-plus-one, $m + 1$ can be absent from ν only if $m + 1$ is a run bottom of T ; but $m + 1 \notin T$. So $m + 1 \notin A$.

Case $m - 1 \notin T$. Then $\{m\}$ is a singleton run of T' and every other run of T' is a run of T . By the first fact $m \in A$; by the second $m + 1 \notin A$; so condition (2) holds for the run $\{m\}$. The blocks of the other runs are unchanged and are disjoint from $[m, m + 1]$: a block $[a, b + 1]$ contains m only if $m \in T$ (no) or $m = b + 1$, i.e. $m - 1 = b \in T$ (excluded in this case); and it contains $m + 1$ only if $m + 1 \in T$ (no) or $m + 1 = b + 1$, i.e. $m \in T$ (no). By Lemma 2.4, $\nu' = (\nu \setminus \{m\}) \cup \{m + 1\}$.

Case $m - 1 \in T$. Let $[a, m - 1]$ be the run of T ending at $m - 1$; the corresponding run of T' is $[a, m]$. Condition (2) for T gives $a \in A$ and $[a + 1, m] \cap A = \emptyset$; with $m + 1 \notin A$ this is exactly (2) for $[a, m]$. All other runs are untouched. By Lemma 2.4 the travelling bead now lands at $m + 1$ instead of m , so again $\nu' = (\nu \setminus \{m\}) \cup \{m + 1\}$.

The second step. $S' = S \setminus \{m\}$ replaces the run $[m, M]$ by $[m + 1, M]$ (empty if $M = m$) and leaves the other runs of S alone. Condition (2) for $[m + 1, M]$ on ν' reads $m + 1 \in \nu'$ — true by construction — and $[m + 2, M + 1] \cap \nu' = \emptyset$, which follows from $(*)$ because ν' differs from ν only at m (removed) and $m + 1$ (added), neither of which lies in $[m + 2, M + 1]$. The other runs of S have blocks disjoint from $[m, M + 1]$, so their conditions on ν' are the same as on ν . By Lemma 2.4 the bead now at $m + 1$ travels to $M + 1$, and every other bead does what it did before; hence $u_{S'} \cdot \nu' = u_S \cdot \nu = \lambda$. $\square$

Corollary 4.2 (a second proof of Theorem 1.1 on the cylindric locus). *For realisable (S, T) , Theorem 4.1 re-proves the identity $u_{S'} u_{T'} = v$ of [ClioExch, Lem. 4.3] by particle bookkeeping alone, with no conjugation computation in $\tilde{S}_N$: the two factorisations move the same beads along the same arcs and differ only in when the arc m is crossed. The original proof is the one that covers the whole of $\tilde{S}_N$; this one covers only the 321-avoiding locus, and is included because it is the reason the statement is true there.*

Corollary 4.3 (the arc multiset is what v remembers). *The multiset $S \uplus T$ of arcs crossed is invariant under the exchange move. Computationally (§6, **V5**) it is constant on the whole fibre $\mathcal{F}(v)$ exactly when v is 321-avoiding: constant on all 1343 321-avoiding fibres for $N \leq 7$, and on no other fibre. This is reported as computed, not proved.*

5 The death of (C)

Theorem 5.1 ((C) is false). *There is no identity between $|\mathcal{E}(S, T)|$ and the number of available hops of the state ν of (3), for two independent reasons.*

(K1) Structural. *By [ClioExch, Thm. 3.1] (localisation), $\mathcal{E}(S, T)$ — indeed the whole fibre — depends only on $v = u_S u_T$. The number of available hops of ν does not: it depends on the base point A . Smallest non-degenerate witness, $N = 4$, $S = \{0\}$, $T = \{1\}$:*

$$A = \{0, 1\} \Rightarrow \nu = \{0, 2\} \text{ with 2 available hops,} \quad A = \{0, 1, 3\} \Rightarrow \nu = \{0, 2, 3\} \text{ with 1.}$$

A quantity that is not a function of v cannot equal one that is. Non-constancy occurs for 1166 of the 4459 realisable pairs with $N \leq 7$.

(K2) Numerical. Over the 9168 realisable triples (S, T, A) with $N \leq 7$, $|\mathcal{E}(S, T)|$ equals the number of available forward hops of ν in 2111 cases; of λ in 2258; of A in 2258. The smallest non-degenerate failure is $N = 3$, $S = \{0\}$, $T = \{1\}$, $A = \{0, 1\}$: no exchange move, one available hop.

By contrast the replacement count (4) holds in 9168 of 9168 cases.

Proof. (K1) is Theorem 1.1’s localisation together with the displayed witness, which is checked directly from (1). (K2) is a finite computation, §6. $\square$

5.1 What survives, and what the right statement is

The mechanism the dream identified is real; the estimand was wrong. The exchange move set is not a set of hops of a *state*. It is a set of re-timings of a two-step *trajectory* whose endpoints, particles and total displacements are already fixed by v . This is exactly why a single-state hop count cannot measure it, and it is visible in Corollary 4.3: what v determines is the multiset of arcs to be crossed; what the fibre $\mathcal{F}(v)$ parametrises is the schedule.

(C’), proved (Theorems 3.1, 4.1). On the 321-avoiding locus, the Theorem 4.1 exchange moves at (S, T) are in bijection with the step-2 journeys whose first two arcs are both free at step 1, and each move advances the intermediate configuration by one TASEP hop of that journey’s particle. The movable arc is always the *first* arc of a journey, never an interior one, because in an exclusion process only the rear of a moving block can be re-timed.

Remark 5.2 (the locality claim, made precise). (C) called the Theorem 4.1 condition “local”. Theorem 3.1 says how local: the usability of a run $[m, M]$ is decided by the two sites m , $m + 1$ alone (“ $m \notin T$ and $m + 1 \notin T$ ”), whereas off the locus the condition genuinely scans the run, and $e > m$ occurs in 2460 of the 6270 non-321-avoiding pairs with $N \leq 7$. So locality is not a property of Theorem 4.1; it is a property of Theorem 4.1 *restricted to the cylindric locus*, and the restriction is what buys it.

5.2 A second negative: the Morse–Schilling excess is not the locus

[ClioCrystal] proved that $\bigcup_x \{\tilde{e}_1^{(x)}(uv)\} \subsetneq \mathcal{E}(S, T)$, strictly, first at $N = 5$, with excess 0, 0, 15, 132, 854 for $N = 3, \dots, 7$ (reproduced here as a positive control, §6, **V1**). It would have been attractive if the excess were exactly the non-cylindric part. It is not: of the 1001 instances with excess, 574 are 321-avoiding (by N : 0, 0, 15, 90, 469). So the strictness of the Morse–Schilling containment is *not* explained by the cylindric/non-cylindric split, and the dream’s suggestion that the strictness “is the price of being a function” is neither confirmed nor refuted here. It remains open.

6 Computational verification

Code at `proofs/code-q227/` (`beads.py`, `order.py`, `lemma1.py`, `verdict.py`, `kill.py`, `invar.py`), running on the existing `proofs/code-q220/` harness. Ranges: all 10729 length-additive pairs for $3 \leq N \leq 7$ (the 9093-instance census of [ClioCrystal] is these minus the 1636 pairs with $X(uv) = \emptyset$), and all 9168 realisable triples (S, T, A) .

- V0** *Positive control, inherited.* `code-q220/final.py` reproduces the published Morse–Schilling excess sequence 0, 0, 15, 132, 854 and the pair counts 34, 143, 561, 2122, 7869. The Morse–Schilling implementation it rests on reproduces all three worked examples printed in [MS]; nothing is fitted.
- V1** *Lemma 2.4, exhaustively.* Predicted nonzero-ness (2) and predicted image checked against the composed action (1) on all 85860 pairs (S, A) with $3 \leq N \leq 8$: 85860 agreements, 0 disagreements. *Could fail:* an off-by-one in the operator order or in the wrap-around would break it immediately.
- V2** *Proposition 2.7.* “realisable” versus “321-avoiding” (tested by the pattern criterion on the window over three periods): agreement 10729/10729.
- V3** *Theorem 3.1.* On all 4459 realisable pairs: $m \notin T$ for every run of S , 4459/4459; $e = m$ for every usable run, 4459/4459; every unusable run satisfies $[m + 1, M + 1] \subseteq T$, 4459/4459; and the count (4), 4459/4459 pairs and 9168/9168 triples. Off the locus, $e > m$ in 2460 of 6270 pairs — so the claim has real content and is not an artefact of the definition.
- V4** *Theorem 4.1 and Corollary 4.2.* For every realisable triple and every exchange move, the image of the full word for $u_{S'}u_{T'}$ on A equals that of u_Su_T : 1326/1326 at $N = 6$ and likewise below, 0 exceptions.
- V5** *Corollary 4.3.* $S \uplus T$ constant on $\mathcal{F}(v)$: 1343/1343 321-avoiding fibres, and 1343 of 3019 fibres in total, the two counts agreeing at every N — so constancy holds exactly on the 321-avoiding fibres.
- V6** *Theorem 5.1.* As displayed in §5.

Remark 6.1 (what the computation could not have caught). Every observable above is computed from (1). If the transport of Lam’s rule into bead coordinates were wrong, **V2** would be the detector: realisability would not track 321-avoidance, and Lam’s dichotomy is sharp enough that a wrong convention would break it. It does not break. This is the same control used in [Clio321, §6], run again here independently.

7 Gaps

- Corollary 4.3 is computed, not proved.** The forward direction (constant on 321-avoiding fibres) should follow from Lemma 2.4 plus the uniqueness statement [Lam, l.1723] (“at most one w with $u_w \cdot \mu = \lambda$ ”), since $S \uplus T$ is then the content multiset of λ/A ; the converse is not attempted.
- The Morse–Schilling excess is unexplained.** §5 rules out the cylindric split as the explanation. The dream’s claim that the strictness is the price of canonicity is untested.
- Semistandard versus standard, and the growth-diagram route.** Nothing here uses [Eliz-A]’s local rules (F1)/(F2) or the correspondence CRSK. The connection asserted in §2 is at the level of the underlying particle model only. A genuine statement relating the exchange move to cylindric growth diagrams would need the rules themselves, and is not claimed.

4. **Novelty.** No claim of novelty is made for the TASEP reading of cylindric tableaux, which is [Eliz-B], nor for the 321-avoiding $\leftrightarrow$ cylindric correspondence, which is [Lam]. What is claimed new is the *route*: the identification of the exchange move of [ClioExch] with a one-arc re-timing, and the consequent proof that $e = m$ on the locus.

References

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